



Evolution

Evolutionary Biology is the study of **history of life forms on earth**.

6.1 ORIGIN OF LIFE

The **origin of life** is considered a **unique event in the history of universe**. The universe is vast and **very old - almost 13.8 billion years old**. Huge **clusters of galaxies** comprise the universe; **galaxies contain stars and clouds of gas and dust**. Considering the size of universe, the **earth is indeed a speck**.

The **Big Bang theory** attempts to explain the **origin of universe**. The sequence from that event to the first oceans on earth runs as follows:

- 1 **Big Bang** - a **singular huge explosion** unimaginable in physical terms. The universe **expanded** and hence the **temperature came down**.
- 2 **Hydrogen and Helium** formed sometime later.
- 3 The gases **condensed under gravitation** and formed the **galaxies** of the present day universe.
- 4 In the **solar system of the milky way galaxy**, **earth** was supposed to have been formed about **4.5 billion years back**.
- 5 There was **no atmosphere on early earth**. **Water vapour, methane, carbon dioxide and ammonia** released from the molten mass covered the surface.
- 6 **UV rays** from the sun **broke up water into Hydrogen and Oxygen**, and the lighter **H₂ escaped**. Oxygen combined with ammonia and methane to form **water, CO₂** and others. The **ozone layer** was formed.
- 7 As it cooled, the **water vapour fell as rain**, to fill all the depressions and **form oceans**.

Life appeared 500 million years after the formation of earth, i.e., almost four billion years back.

6.1.1 Theories on the Origin of Life

6.1.1.1 Panspermia

Some scientists believe that it came from outside. **Early Greek thinkers** thought units of life called **spores** were transferred to different planets including earth. **Panspermia** is still a favourite idea for some **astronomers**.

6.1.1.2 Spontaneous Generation

For a long time it was also believed that life came out of **decaying and rotting matter like straw, mud**, etc. This was the **theory of spontaneous generation**.

Louis Pasteur, by careful experimentation, demonstrated that **life comes only from pre-existing life**. He showed that in **pre-sterilised flasks**, life did **not** come from **killed yeast**, while in another flask **open to air**, new living organisms arose from killed yeast. **Spontaneous generation theory was dismissed once and for all**. However, this did **not** answer how the **first life form** came on earth.

6.1.1.3 Chemical Evolution

Oparin of Russia and **Haldane of England** proposed that the first form of life could have come from **pre-existing non-living organic molecules** (for example **RNA and protein**), and that formation of life was **preceded by chemical evolution**.

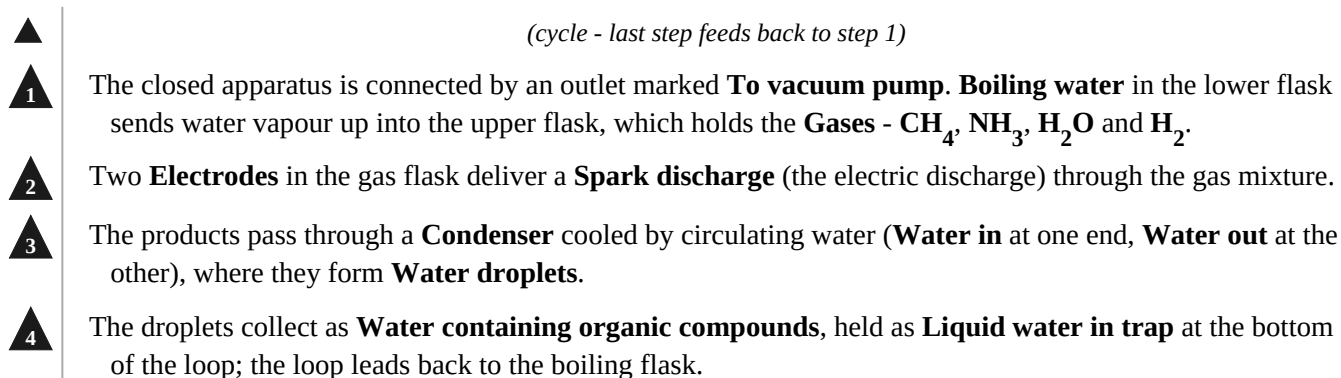
● **Chemical evolution** - the **formation of diverse organic molecules from inorganic constituents**.

The conditions on earth then were **high temperature, volcanic storms**, and a **reducing atmosphere** containing **CH₄, NH₃**, etc.

In 1953, **S. L. Miller**, an **American scientist**, created similar conditions on a laboratory scale. He created **electric discharge** in a **closed flask** containing **CH₄, H₂, NH₃** and **water vapour at 800 degrees C**, and observed the **formation of amino acids**. In similar experiments others observed formation of **sugars, nitrogen bases, pigment**

and fats.

Miller's experiment (NCERT Figure 6.1 - diagrammatic representation of Miller's experiment). The apparatus works as a closed loop:



Analysis of meteorite content also revealed similar compounds, indicating that similar processes are occurring **elsewhere in space**. With this limited evidence, the first part of the conjectured story, i.e., **chemical evolution, was more or less accepted**.

We have **no idea** about how the **first self-replicating metabolic capsule of life** arose. The **first non-cellular forms of life** could have originated **3 billion years back**. They would have been **giant molecules**, such as **RNA, protein, polysaccharides**, etc. These capsules reproduced their molecules perhaps.

The **first cellular form of life** did not possibly originate till about **2000 million years ago**; these were probably **single-cells**. **All life forms were in water environment only**. This version of a **biogenesis**, i.e., that the first form of life arose slowly through evolutionary forces from non-living molecules, is **accepted by majority**.

6.2

EVOLUTION OF LIFE FORMS - A THEORY

Conventional religious literature tells us about the **theory of special creation**. This theory has **three connotations**:

- **One** - all living organisms (species or types) that we see today were **created as such**.
- **Two** - the **diversity was always the same** since creation and will be the same in future also.
- **Three** - **earth is about 4000 years old**.

All these ideas were **strongly challenged during the nineteenth century**.

Based on observations made during a sea voyage in a sail ship called **H.M.S. Beagle** round the world, **Charles Darwin** concluded that existing living forms **share similarities to varying degrees** not only among themselves but also with life forms that existed **millions of years ago**. Many such life forms **do not exist any more**: there had been **extinctions** of different life forms in the years gone by, just as new forms of life arose at different periods of the history of earth. There has been **gradual evolution of life forms**.

Any population has built-in variation in characteristics. Those characteristics which enable some to **survive better in natural conditions** (such as **climate, food and physical factors**) would **outbreed** others that are less endowed to survive under such natural conditions.

- **Fitness** of the individual or population - according to Darwin, it refers **ultimately and only to reproductive fitness**.
- **Natural selection** - those who are **better fit** in an environment **leave more progeny** than others; these therefore survive more and are **selected by nature**. Darwin called it natural selection and implied it as a **mechanism of evolution**.

Alfred Wallace, a naturalist who worked in the **Malay Archipelago**, had also come to **similar conclusions around the same time**.

In due course of time, it became apparent that **all the existing life forms share similarities and share common ancestors**. These ancestors were present at different periods in the history of earth - **epochs, periods and eras**. The **geological history of earth closely correlates with the biological history of earth**. A common conclusion is that earth is very old - **not thousands of years** as was thought earlier **but billions of years old**.

Evidence that evolution of life forms has indeed taken place on earth has come from **many quarters**.

Paleontological Evidence

Fossils are **remains of hard parts of life-forms found in rocks**. Rocks form **sediments**, and a cross-section of earth's crust indicates the arrangement of sediments **one over the other** during the long history of earth. **Different-aged rock sediments contain fossils of different life-forms**, who probably died during the formation of the particular sediment. Some of them appear similar to modern organisms; they represent **extinct organisms** (for example **dinosaurs**).

A study of fossils in different sedimentary layers indicates the **geological period** in which they existed. The study showed that life-forms varied over time and that **certain life forms are restricted to certain geological time-spans**. All this is called **paleontological evidence**.

☆ **[MEMORY AID - not in NCERT]** Radiometric dating estimates fossil ages by comparing a radioactive parent isotope with its stable daughter isotope and using the known half-life to calculate elapsed time.

A **family tree of dinosaurs** and their living modern day counterpart organisms **like crocodiles and birds** (NCERT Figure 6.2) places these forms on one tree: **Triceratops, Tyrannosaurus, Stegosaurus, Brachiosaurus** and **Pteranodon**, alongside the **Crocodylian** line (today's crocodiles) and **Archaeopteryx** on the line leading to birds.

Embryological Support

Embryological support for evolution was proposed by **Ernst Haeckel**, based upon the observation of certain features during **embryonic stage common to all vertebrates** that are **absent in adult**. For example, the embryos of all vertebrates **including human** develop a row of **vestigial gill slit just behind the head**, but it is a functional organ **only in fish** and not found in any other adult vertebrates.

However, this proposal was **disapproved** on careful study performed by **Karl Ernst von Baer**. He noted that **embryos never pass through the adult stages of other animals**.

Comparative Anatomy and Morphology

Comparative anatomy and morphology shows **similarities and differences** among organisms of today and those that existed years ago. Such similarities can be interpreted to understand **whether common ancestors were shared or not**.

For example, **whales, bats, cheetah and human** (all mammals) share similarities in the **pattern of bones of forelimbs**. Though these forelimbs perform **different functions** in these animals, they have **similar anatomical structure** - all of them have **humerus, radius, ulna, carpals, metacarpals and phalanges**.

- **Divergent evolution / homology** - the **same structure developed along different directions due to adaptations to different needs**. These structures are **homologous**; **homology indicates common ancestry** (it is accounted for by the idea of **branching descent**). Other examples: **vertebrate hearts or brains**; in plants, the **thorn and tendrils** of *Bougainvillea* and *Cucurbita*.

Example of homologous organs in (a) Plants and (b) Animals (NCERT Figure 6.3): in plants, the **Thorn** of *Bougainvillea* and the **Tendrils** of *Cucurbita*; in animals, the forelimbs of **Man, Cheetah, Whale and Bat**.

Homology is based on divergent evolution whereas analogy refers to a situation exactly opposite. Wings of butterfly and of birds look alike; they are **not anatomically similar structures** though they perform **similar functions**.

Thus **analogous structures** are a result of **convergent evolution** - **different structures evolving for the same function and hence having similarity**. It is the **similar habitat** that has resulted in selection of similar adaptive features in different groups of organisms, but toward the same function.

Feature	Homologous organs	Analogous organs
Type of evolution	Divergent evolution	Convergent evolution
Anatomy	Similar anatomical structure	Not anatomically similar structures
Function	Different functions (different needs)	Similar functions

Feature	Homologous organs	Analogous organs
What it indicates	Common ancestry	Selection of similar adaptive features by a similar habitat
Animal examples	Forelimbs of whales, bats, cheetah and human ; vertebrate hearts or brains	Wings of butterfly and of birds; eye of the octopus and of mammals; flippers of Penguins and Dolphins
Plant examples	Thorn of Bougainvillea and tendrils of Cucurbita	Sweet potato (a root modification) and potato (a stem modification)

Similarly, **similarities in proteins and genes** performing a given function among diverse organisms give clues to **common ancestry**. These **biochemical similarities** point to the same shared ancestry as **structural similarities** among diverse organisms.

Artificial selection for agriculture, horticulture, sport or security has produced new breeds of domesticated plants and animals, such as dogs, within hundreds of years; natural selection could produce comparable changes over millions of years.

6.3.4 Moths before and after industrialisation

Collection	Tree trunks	Moth that survived	Result
1850s - before industrialisation set in	Covered by thick growth of almost white-coloured lichen	White-winged moth ; dark-coloured moths were picked out by predators	More white-winged moths on trees than dark-winged or melanised moths
1920 - after industrialisation, same area	Became dark due to industrial smoke and soots	Dark-winged or melanised moth ; the white-winged moth did not survive due to predators	More dark-winged moths - the proportion was reversed

NCERT Figure 6.4 shows the **white-winged moth and dark-winged moth (melanised) on a tree trunk (a) in unpolluted area (b) in polluted area**: on the pale, lichen-covered trunk the dark moth stands out, while on the soot-darkened trunk the white moth stands out. The explanation is that **predators will spot a moth against a contrasting background**; moths that were able to **camouflage** themselves, i.e., **hide in the background**, survived. Supporting this, in **rural areas where industrialisation did not occur**, the **count of melanic moths was low**. This showed that in a **mixed population**, those that can better adapt survive and increase in population size. Remember that **no variant is completely wiped out**.

① [NOTE] Lichens can be used as industrial pollution indicators - they will not grow in areas that are polluted.

Similarly, **excess use of herbicides, pesticides, etc.**, has only resulted in **selection of resistant varieties** in a **much lesser time scale**. This is also true for **microbes against which we employ antibiotics** or drugs against eukaryotic organisms or cells. Hence, **resistant organisms or cells** are appearing in a time scale of **months or years and not centuries**. These are examples of **evolution by anthropogenic action**.

This also tells us that **evolution is not a directed process**, but is a **stochastic process** based on **chance events in nature** and **chance mutation** in the organisms.

6.4 WHAT IS ADAPTIVE RADIATION?

During his journey **Darwin went to Galapagos Islands**. There he observed an amazing diversity of creatures. Of particular interest, small black birds later called **Darwin's Finches** amazed him: he realised that there were **many varieties of finches in the same island**. All the varieties, he conjectured, **evolved on the island itself**. From the original **seed-eating** features, many other forms with **altered beaks** arose, enabling them to become **insectivorous and vegetarian finches**. NCERT Figure 6.5 shows this **variety of beaks of finches that Darwin found in Galapagos Island** - four beak forms side by side.

- **Adaptive radiation** - the process of evolution of **different species in a given geographical area** starting from a point and literally **radiating to other areas of geography, or habitats**. **Darwin's finches** represent one of the best examples.

Another example is **Australian marsupials**: a number of marsupials, each different from the other, evolved from an **ancestral stock**, but **all within the Australian island continent**. NCERT Figure 6.6 (**adaptive radiation of marsupials of Australia**) shows this **Marsupial radiation** fanning out across **Australia** from one ancestral stock into these marsupials, each different from the other:

- **Sugar glider, Tasmanian wolf, Tiger cat, Marsupial mole, Banded anteater, Koala, Marsupial rat, Bandicoot, Wombat and Kangaroo.**

When **more than one adaptive radiation** appeared to have occurred in an **isolated geographical area** (representing different habitats), one can call this **convergent evolution**. **Placental mammals in Australia** also exhibit adaptive radiation in evolving into varieties of such placental mammals, **each of which appears to be similar to a corresponding marsupial** - for example **placental wolf and Tasmanian wolf-marsupial**. NCERT Figure 6.7 (**convergent evolution of Australian marsupials and placental mammals**) pairs them as follows:

Placental mammals	Australian marsupials
Mole	Marsupial mole
Anteater	Numbat (anteater)
Mouse	Marsupial mouse
Lemur	Spotted cuscus
Flying squirrel	Flying phalanger
Bobcat	Tasmanian tiger cat
Wolf	Tasmanian wolf

6.5

BIOLOGICAL EVOLUTION

Evolution by natural selection, in a true sense, would have started when **cellular forms of life with differences in metabolic capability** originated on earth. The **essence of Darwinian theory** about evolution is **natural selection**. The **rate of appearance of new forms is linked to the life cycle or the life span**.

Microbes that divide fast have the ability to multiply and become millions of individuals **within hours**. A colony of bacteria (say **A**) growing on a given medium has built-in variation in terms of ability to utilise a feed component. A change in the medium composition would bring out only that part of the population (say **B**) that can survive under the new conditions. In due course of time this **variant population outgrows the others and appears as new species**. This would happen **within days**. For the same thing to happen in a **fish or fowl** would take **millions of years**, as life spans of these animals are in years. We say that the **fitness of B is better than that of A** under the new conditions. **Nature selects for fitness**.

The so-called fitness is based on characteristics which are **inherited**; hence there must be a **genetic basis** for getting selected and to evolve. In other words, **adaptive ability is inherited**; it has a genetic basis. **Fitness is the end result of the ability to adapt and get selected by nature**.

● **Branching descent and natural selection** are the **two key concepts of Darwinian Theory of Evolution**.

Even before Darwin, a French naturalist **Lamarck** had said that evolution of life forms had occurred but **driven by use and disuse of organs**. His example was the **giraffe**: in an attempt to forage leaves on tall trees, giraffes had to adapt by **elongation of their necks**; as they passed on this **acquired character** of elongated neck to succeeding generations, giraffes slowly, over the years, came to acquire long necks. **Nobody believes this conjecture any more**.

It is possible that the work of **Thomas Malthus on populations** influenced Darwin. Natural selection is based on certain observations which are factual; Darwin's reasoning runs step by step:

1

Natural resources are limited.

2

Populations are stable in size except for **seasonal fluctuation**.

3

Members of a population vary in characteristics - in fact **no two individuals are alike** - even though they look superficially similar.

4

Most variations are inherited.

5

Population size will **grow exponentially** if everybody reproduced maximally; yet real population sizes are limited. This means there had been **competition for resources**: only some survived and grew at the cost of others that could not flourish.

6

Darwin's insight: **variations which are heritable** and which make **resource utilisation better for few** (adapted to habitat better) will enable **only those to reproduce and leave more progeny**.

7

Over many generations the **survivors leave more progeny**, there is a **change in population characteristic**, and hence **new forms appear to arise**.

Operation of natural selection on different traits (NCERT Figure 6.8 - (a) **Stabilising**, (b) **Directional** and (c) **Disruptive**). Each panel plots the **Number of individuals with phenotype** against the phenotype, with the shaded part marking the **Phenotypes favoured by natural selection**:

Mode of selection	Which phenotypes are favoured	Change in the curve
(a) Stabilising	Medium-sized individuals are favoured	Peak gets higher and narrower
(b) Directional	Individuals at one end of the distribution	Peak shifts in one direction
(c) Disruptive	Individuals at both ends (peripheral values)	Two peaks form

6.6

MECHANISM OF EVOLUTION

Darwin did not explain the origin of variation or speciation; although Mendel described inheritable factors influencing phenotype, Darwin ignored those observations or remained silent.

In the **first decade of twentieth century**, **Hugo de Vries**, based on his work on **evening primrose**, brought forth the idea of **mutations - large difference arising suddenly in a population**. He believed that it is **mutation which causes evolution**, and not the minor variations (heritable) that Darwin talked about.

Point	Darwin	Hugo de Vries
Cause of evolution	Minor, heritable variations	Mutation
Nature of the change	Variations are small and directional	Mutations are random and directionless
Pace	Evolution was gradual	Mutation caused speciation in one step - saltation (single-step large mutation)

Studies in **population genetics**, later, brought out some clarity.

6.7

HARDY-WEINBERG PRINCIPLE

In a given population one can find out the **frequency of occurrence of alleles of a gene or a locus**. This frequency is supposed to **remain fixed** and even remain the same through generations. The **Hardy-Weinberg principle** stated it using algebraic equations: **allele frequencies in a population are stable and are constant from generation to generation**.

The **gene pool** - the total genes and their alleles in a population - **remains a constant**. This is called **genetic equilibrium**.

Sum total of all the allelic frequencies is 1. Individual frequencies can be named **p, q**, etc. In a **diploid**, **p** and **q** represent the frequency of allele **A** and allele **a**. The frequency of **AA** individuals in a population is simply **p²** - the probability that an allele **A** with frequency **p** appears on **both** the chromosomes of a diploid individual is the product of the probabilities, i.e., **p x p = p²**. Similarly the frequency of **aa** is **q²**, and that of **Aa** is **2pq**. Hence:

$$p^2 + 2pq + q^2 = 1$$

This is a **binomial expansion of (p + q)²**.

When the **frequency measured differs from expected values**, the difference (or direction) indicates the **extent of evolutionary change**. Disturbance in genetic (Hardy-Weinberg) equilibrium, i.e., **change of allele frequency** in a population, would then be interpreted as **resulting in evolution**.

Five factors are known to affect Hardy-Weinberg equilibrium: **gene migration or gene flow, genetic drift, mutation, genetic recombination and natural selection.**

Factor	How it changes allele frequency
Gene migration / gene flow	When a section of population migrates to another place and population, gene frequencies change in the original as well as in the new population : new genes or alleles are added to the new population and lost from the old . There would be a gene flow if this gene migration happens multiple times .
Genetic drift	If the same change occurs by chance , it is called genetic drift. Sometimes the change in allele frequency is so different in the new sample of population that they become a different species ; the original drifted population becomes founders and the effect is called founder effect .
Mutation	Microbial experiments show that pre-existing advantageous mutations when selected will result in observation of new phenotypes ; over few generations this would result in speciation .
Genetic recombination	Variation due to recombination during gametogenesis changes the frequency of genes and alleles in future generations.
Natural selection	A process in which heritable variations enabling better survival are enabled to reproduce and leave greater number of progeny .

- **Genetic drift** - a change in allele frequency that occurs **by chance**; the drifted population that becomes a different species acts as **founders (founder effect)**.

Variation due to **mutation**, or variation due to **recombination during gametogenesis**, or due to **gene flow or genetic drift**, results in **changed frequency of genes and alleles** in future generation. Coupled to **enhanced reproductive success**, natural selection makes it look like a **different population**.

- **Habitat fragmentation** - other phenomena like **habitat fragmentation and genetic drift** may **accentuate these variations**, leading to **appearance of new species** and hence **evolution**.

Natural selection can lead to:

- **Stabilisation** - more individuals acquire mean character value.
- **Directional change** - more individuals acquire value other than the mean character value.
- **Disruption** - more individuals acquire peripheral character value at both ends of the distribution curve.

☆ **[MEMORY AID - not in NCERT]** "*Stay, Drive, Split*" - *Stabilising keeps the middle (one peak, higher and narrower); Directional drives the peak to one side; Disruptive splits the curve into two peaks.*

6.8

A BRIEF ACCOUNT OF EVOLUTION

About **2000 million years ago (mya)** the **first cellular forms of life** appeared on earth. The mechanism of how **non-cellular aggregates of giant macromolecules** could evolve into cells with **membranous envelop** is **not known**. Some of these cells had the ability to **release O₂**; the reaction could have been similar to the **light reaction in photosynthesis**, where water is split with the help of solar energy captured and channelised by appropriate **light harvesting pigments**.

When (as stated)	Event
about 2000 mya	First cellular forms of life appeared
by the time of 500 mya	Slowly single-celled organisms became multi-cellular life forms; invertebrates were formed and active
probably around 350 mya	Jawless fish evolved
probably around 320 mya	Sea weeds and few plants existed
about 350 mya	Fish with stout and strong fins could move on land and go back to water (lobefins)
next 200 million years or so	Reptiles of different shapes and sizes dominated on earth
probably 200 mya	Some land reptiles went back into water to evolve into fish-like reptiles (for example <i>Ichthyosaurs</i>)
about 65 mya	The dinosaurs suddenly disappeared from the earth

We are told that the **first organisms that invaded land were plants**; they were **widespread on land when animals invaded land**. In 1938, a fish caught in **South Africa** happened to be a *Coelacanth*, which was thought to be **extinct**. These animals, called **lobefins**, evolved into the **first amphibians** that lived on both land and water. There are **no specimens** of these left with us; however, these were **ancestors of modern day frogs and salamanders**.

The **amphibians evolved into reptiles**. Reptiles lay **thick-shelled eggs which do not dry up in sun unlike those of amphibians**. Again we only see their modern day descendents - the **turtles, tortoises and crocodiles**. **Giant ferns (pteridophytes)** were present, but they all fell to form **coal deposits** slowly.

The land reptiles were, of course, the **dinosaurs**. The biggest of them, i.e., *Tyrannosaurus rex*, was **about 20 feet in height** and had huge fearsome **dagger-like teeth**. About 65 mya the dinosaurs suddenly disappeared - **we do not know the true reason**. Some say **climatic changes killed them**; some say **most of them evolved into birds**. **Small-sized reptiles of that era still exist today**.

A rough sketch of the evolution of life forms and their times on a geological scale is given in NCERT Figures 6.9 and 6.10. **A sketch of the evolution of plant forms through geological periods** (Figure 6.9) is read as two tables - first the time scale, then the plant groups:

Era	Periods (oldest first)
Paleozoic	Silurian, Devonian, Carboniferous, Permian
Mesozoic	Triassic, Jurassic, Cretaceous
Cenozoic	Tertiary, Quaternary

Part of the sketch	Plant forms shown
Base of the tree (ancestral stock)	chlorophyte ancestors , then tracheophyte ancestors , Rhynia-type plants and Psilophyton (Silurian to Devonian)
Early side lines	Bryophytes (from the chlorophyte ancestors); Zosterophyllum , leading to herbaceous lycopods and arborescent lycopods
Lines arising from Psilophyton	Sphenopsids (horsetails) , Ferns and Progymnosperms
Seed plants	Progymnosperms give seed ferns ; the seed-plant lines include Conifers , Cycads , Ginkgos and Gnetales
Youngest line (Cretaceous onwards)	Angiosperms (flowering plants) , branching into Monocotyledons and Dicotyledons

Representative evolutionary history of vertebrates through geological periods (Figure 6.10) runs from the **Carboniferous** through the **Permian, Triassic, Jurassic, Cretaceous** and **Tertiary** to the **Quaternary**. From **Early reptiles (extinct)** at the base, two lines split:

Line	Forms along the line	Living descendants
Sauropsids	Thecodonts (extinct) , which gave rise to Dinosaurs (extinct)	Turtles, Lizards, Snakes, Tuataras, Crocodiles, Birds
Synapsids	Pelycosaurs (extinct) , then Therapsids (extinct)	Mammals

The **first mammals were like shrews**; their fossils are **small sized**. Mammals were **viviparous** and protected their unborn young inside the mother's body. Mammals were **more intelligent in sensing and avoiding danger** at least. When reptiles came down, **mammals took over this earth**.

There were in **South America** mammals resembling **horse, hippopotamus, bear, rabbit**, etc. Due to **continental drift**, when South America joined North America, these animals were **overridden by North American fauna**. Due to the same continental drift, **pouched mammals of Australia survived** because of **lack of competition** from any other mammal. Some mammals **live wholly in water** - **whales, dolphins, seals and sea cows** are some examples.

Evolution of horse, elephant, dog, etc., are special stories of evolution. The **most successful story is the evolution of man with language skills and self-consciousness**.

About **15 mya**, primates called *Dryopithecus* and *Ramapithecus* were existing. They were **hairy and walked like gorillas and chimpanzees**. *Ramapithecus* was **more man-like** while *Dryopithecus* was **more ape-like**. Few fossils of man-like bones have been discovered in **Ethiopia and Tanzania**; these revealed **hominid features**, leading to the belief that about **3-4 mya**, man-like primates walked in **eastern Africa**. They were probably **not taller than 4 feet** but **walked upright**.

When (as stated)	Form	Key features (NCERT)
about 15 mya	<i>Dryopithecus</i> and <i>Ramapithecus</i>	Hairy; walked like gorillas and chimpanzees; <i>Ramapithecus</i> more man-like, <i>Dryopithecus</i> more ape-like
about 3-4 mya	Man-like primates of eastern Africa	Probably not taller than 4 feet ; walked upright
Two mya	Australopithecines	Probably lived in East African grasslands ; hunted with stone weapons but essentially ate fruit
-	<i>Homo habilis</i> - the first human-like being, the hominid	Brain capacities between 650-800 cc ; probably did not eat meat
about 1.5 mya	<i>Homo erectus</i>	Fossils discovered in Java in 1891 ; large brain around 900 cc ; probably ate meat
100,000-40,000 years back	Neanderthal man	Brain size 1400 cc ; lived in Near East and central Asia ; used hides to protect their body and buried their dead
ice age, 75,000-10,000 years ago	Modern <i>Homo sapiens</i>	<i>Homo sapiens</i> arose in Africa , moved across continents and developed into distinct races ; modern <i>Homo sapiens</i> arose during this ice age

NCERT Figure 6.11 makes a **comparison of the skulls of adult modern human being, baby chimpanzee and adult chimpanzee**: the skull of baby chimpanzee is more like adult human skull than adult chimpanzee skull.

Pre-historic cave art developed about **18,000 years ago**. One such cave painting by pre-historic humans can be seen at **Bhimbetka rock shelter in Raisen district of Madhya Pradesh**. **Agriculture came around 10,000 years back** and **human settlements started**. The rest of what happened is part of human history of growth and decline of civilisations.

☆ **[MEMORY AID - not in NCERT]** Brain size climbs in hundreds: *Homo habilis* 650-800, *Homo erectus* about 900, *Neanderthal* 1400 (all in cc) - "**H-E-N**: Habilis, Erectus, Neanderthal" in order of increasing brain size.

Recap

QUICK RECAP

- **Origin of life** makes sense only against the origin of the **universe (about 13.8 billion years)** and of the **earth (about 4.5 billion years)**; life appeared almost **four billion years back**.
- Most scientists hold that **chemical evolution** (formation of biomolecules from inorganic constituents - Oparin, Haldane; Miller 1953) **preceded the first cellular forms of life** (about 2000 mya). Pasteur dismissed **spontaneous generation**.
- What followed is a **conjectured story** built on **Darwinian ideas of organic evolution by natural selection**: **branching descent** and **natural selection** are its two key concepts; fitness means **reproductive fitness**.
- Diversity of life forms has been **changing over millions of years**. **Variations in a population result in variable fitness**; **habitat fragmentation and genetic drift** may accentuate these variations, leading to **new species** and hence evolution.
- Evidence for evolution comes from **fossils (paleontology)**, **comparative anatomy** (homology = divergent evolution, accounted for by **branching descent**; analogy = convergent evolution) and **comparative biochemistry**, plus industrial melanism and anthropogenic resistance.
- **Adaptive radiation**: Darwin's finches, Australian marsupials; several radiations in one isolated area give **convergent evolution** (placental vs marsupial look-alikes).
- **Hardy-Weinberg**: $p^2 + 2pq + q^2 = 1$; disturbed by **gene flow, genetic drift, mutation, genetic recombination and natural selection**. Selection may be **stabilising, directional or disruptive**. de Vries: **mutation, saltation**.

- The story of **modern man** is the most interesting, and appears to **parallel the evolution of human brain and language** - from *Dryopithecus/Ramapithecus* through *Homo habilis* (650-800 cc), *Homo erectus* (900 cc) and Neanderthal (1400 cc) to *Homo sapiens*.

Appendix

TERMS USED IN THE EXERCISES

Exercise 3 asks for a **clear definition of the term species**. The chapter uses the word throughout but never defines it; the definition below is an **addition, not stated in this NCERT chapter**.

- **Species** - a group of individuals that are **similar in their characteristics** and can **actually or potentially interbreed in nature to produce fertile offspring**, while being **reproductively isolated** from other such groups (the **biological species concept**).